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SARDAR PATEL UNIVERSITY

B. Sc. Examination (Fifth semester)

21-8-2019 : 10 to 12 PM

US05EICH02 (Industrial Chemistry-I)

Total Marks: 70

Q-1 Choose the most appropriate option for each of the following. [10]

1. _____ balance that indicate what is happening in a system at an instant of time.
(a) Differentia (b) Material (c) Integral (d) Energy
2. A material balance is nothing more than an accounting for _____ and changes in inventory of material for a system.
(a) volumetric flows (b) mass flows (c) input flows (d) material flows
3. In case of _____ fins, long metal strips are attached to the outside of the pipe.
(a) transverse (b) disc type (c) helical fins (d) Longitudinal
4. The spur gear pump contains _____ gear wheels in casing.
(a) one (b) four (c) two (d) six
5. For external cleaning of the tubes, tube bundle can be taken out of shell in _____ heat exchanger.
(a) floating head (b) fixed tube sheet (c) U-tube type (d) kettle type
6. Thermometers are employed for the measurement of _____.
(a) pressure (b) temperature (c) volume (d) energy
7. Mercury-in-glass thermometer cannot be used below _____ °F.
(a) 27 (b) 38 (c) 50 (d) 72
8. 136 mg of CaSO_4 in 1000 ml of water is equivalent to CaCO_3
(a) 100 ppm (b) 150 ppm (c) 50 ppm (d) 75 ppm
9. Carry over of steam mainly due to _____.
(a) Corrosion (b) Scale Formation (c) Priming & Foaming (d) Caustic embrittlement
10. Iron and Manganese are present in water as _____.
(a) Hydroxide (b) Carbonate (c) Sulphate (4) Chloride

Q-2 Attempt any ten questions of following. [20]

1. Define the following terms: (i) Limiting reactant (ii) Selectivity
2. What is by-passing streams.
3. Batch processes are unsteady-state operations. Explain.
4. Give the types of reciprocating pumps.
5. What is priming?
6. Give the various losses occurring during the operation of a centrifugal pump.

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7. Give the methods of selecting lead wires.
8. Give the advantage and disadvantage of bimetallic thermometer.
9. Explain about the different types of bulbs used in industrial pressure spring thermometer.
10. What are the common impurities present in natural water?
11. Give the methods of boiler water treatment.
12. Give the treatments used for feed water conditioning.

Q-3 Write a note on Recycle operation and its importance. [10]

OR

Q-3 Write a note on Material balance problem and its classification. [10]

Q-4 Attempt the following.

- (a) Give the comparison of centrifugal pump with reciprocating pump. [05]
- (b) Write in brief on plate type heat exchanger. [05]

OR

Q-4 Attempt the following.

- (a) Write notes on Gear pump. [05]
- (b) Write in brief on finned tube heat exchanger. [05]

Q-5 Attempt the following.

- (a) Write notes on mercury in-glass thermometer. [05]
- (b) Give the brief account on industrial thermocouples. [05]

OR

Q-5 Attempt the following.

- (a) Write notes on temperature scale. [05]
- (b) Write notes on radiation pyrometers. [05]

Q-6 Attempt the following.

- (a) Discuss the Sources of water. [05]
- (b) Write note on: Determination of hardness. [05]

OR

Q-6 Attempt the following.

- (a) Write notes on blow down. [05]
- (b) Calculate the amount of Lime-soda needed for softening water containing the following per liter: [05]
 $\text{Ca}(\text{HCO}_3)_2$: 162 mg; $\text{Mg}(\text{HCO}_3)_2$: 73 mg; MgCl_2 : 95 mg; NaCl : 585 mg;
 CaSO_4 : 136 mg.

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